Emerging success of flexible strategies in increasingly noisy Iterated Prisoner's Dilemma

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Abstract
The importance of game theory in the modern world of evolving politics cannot be overstated. The Iterated Prisoner's Dilemma(IPD) is a simulation of interactions between two entities, called strategies, with choices to cooperate or defect. It is a useful tool for simulating international conflicts and studying strategic behaviors. While the original tournaments by Professor Axelrod were a foundational effort, they did not consider noise, which distorts the intentions of each strategy, changing cooperation to defection and vice versa. In real-world politics and international relations, it is inevitable that there be misunderstandings between two parties, and noise is therefore a crucial component when simulating with the IPD. This study aimed to investigate the performance of strategies from Professor Axelrod's first tournament in a noisy environment. Additionally, this research aimed to provide evidence supporting the emergence of a new characteristic of successful strategies in noisy environments, named "flexibility." In this study, the first tournament was recreated in Python, and noise levels were applied in gradually increasing magnitude from 0% to 9% to simulate varying degrees of uncertainty and observe any persisting behaviors that contribute to strategic success. Downing, a strategy that best displays this characteristic, became the consistent top-scorer of the noisy IPD, while strategies that did not display this characteristic scored consistently low. These findings suggest that flexibility is a valid criterion for predicting a strategy's success in noisy environments.

Keywords:
Noise, Flexibility, Game Theory, Iterated Prisoner's Dilemma, Axelrod's Tournaments, Downing, Tit for Tat

1. Introduction

The Iterated Prisoner's Dilemma (IPD) is a concept within the field of game theory that models interactions between two rational agents. In the original Prisoner's Dilemma, the two agents each are given two choices: to cooperate or to defect. If both agents cooperate, they are given the Reward ("R"), in which each agent gains three points. If only one agent cooperates while the other defects, the cooperator is given the Sucker's Payoff ("S"), which is zero points, and the defector gains the Temptation ("T"), which is five points. If both agents resort to defections, they are both given Punishment ("P"), which is one point for each agent. The IPD takes the same metric, but creates a round-robin tournament, with multiple computer-programmed agents called strategies playing multiple rounds of the Prisoner's Dilemma against each other, and the aim is to gain as much payoff as possible. The purpose of studying the IPD is to observe occurrences of characteristics that are consistently successful in the iterated tournaments. (1)

In this study, an additional component, noise, was incorporated into the tournament. Noise is the random errors that occur within the tournament, which inverts the choice of a strategy. When noise is present, a choice of cooperation would be recorded in the tournament and perceived by the opponent as a defection, and vice versa. Noise is controlled by the noise level, which is a number between 0 and 1 that determines the probability of noise. A tournament with a noise level of 0.01 denotes that each choice made by the strategy has 1% chance of being inverted by the noise. These inverted cooperations and defections will be called noisy cooperations and defections. Under these conditions, this study aimed to determine a common characteristic of successful strategies, and compare their performances in noisy and noiseless tournaments.

2. Literature Review

2.1 History of IPD

The IPD has been used extensively to simulate the emergence of diverse behaviors in international conflicts. A historical example is the United States and the Soviet Union's nuclear arms race. According to the prisoner's dilemma, their actions were equivalent to defections, meaning they continued to produce for their nuclear arsenal without adhering to previously agreed nuclear arms reduction. This mutual defection caused continued defection from both sides due to the risk of lacking nuclear power when needed, or receiving the Sucker's Payoff. Both sides ended up in a suboptimal situation, devoting large portions of their budget to nuclear arms development. (7)

2.2 Nash Equilibrium

As discussed, in a single prisoner's dilemma, the best action for a player is to defect. It minimizes the risk while maximizing the potential gain. However, if both players are rational, the opponent will also arrive at the same conclusion, meaning that the result will be mutual defection. This is the Nash equilibrium for the single Prisoner's Dilemma; no other strategy will benefit the player more. This was what was expected of Axelrod's first and second tournaments. As the IPD is a lengthened version of the Prisoner's Dilemma, it was expected that strategies that continuously defect would prevail. However, Professor Axelrod's study found contradictory results.

2.3 Axelrod's Tournaments

Axelrod's First and Second Tournaments have been a foundational study in IPD, discovering four key characteristics of a successful strategy: being nice, forgiving, provokable, and clear. Nice strategies never defected first, while forgiving strategies accepted apologies, such as the opponent cooperating after a defection. (1) Provokable strategies immediately retaliated against defection with defection, and clear strategies made consistent choices that could be interpreted by the opponent. (2) The simplest strategy, named Tit for Tat, won both tournaments, and it embodied all four qualities listed above. Tit for Tat's entire logic was to start by cooperating, then copy the opponent's last move, so it was an unexpected result for Professor Axelrod. This result was significant as it showed that even among rational, self-interested agents, cooperation can emerge. Yet the original research did not incorporate noise: errors occurring within the tournaments that mislead the intentions of the strategy to its opponent. This setup of a noiseless environment introduces limitations for real-world applications.

2.4 Research into Noisy IPD

Continued research on the IPD with noise incorporated found additional characteristics of generosity and contrition, which assist strategies in avoiding defection chains provoked by a noise-caused defection (noisy defections). In this original noisy tournament, Tit for Tat's payoff decreased, due to a chain of defections caused by a noisy defection. (10) For example, if two Tit for Tat players played each other in a noiseless tournament, they would achieve a perfect score for total cooperation. However, if noise was incorporated and a noisy defection occurred, the opponent of the defecting Tit for Tat would retaliate with a defection, and the defecting Tit for Tat would retaliate again, going on to a chain of alternating defections.

The Generous Tit for Tat attempts to resolve this issue by having 10% chance of cooperating even after the opponent has defected. This allows Generous Tit for Tat to break out of the defection chain. Contrite Tit for Tat, on the other hand, seeks to prevent chains of defection by basing its current move not only on the opponent's previous move but also on its own previous move. If they recognize a noisy defection from their own move, they cooperate until the cooperation is reciprocated.

Though generosity and contrition were shown to be successful in certain conditions, they both contain significant weaknesses that prevent their success in all noisy tournaments. Generosity has a high risk of being taken advantage of, since it allows some defections to be ignored. (3) Contrition, on the other hand, can only correct its own noisy defections. It would detect noise in its own move history and apologize, but it is unable to respond effectively to noisy defections from the opponent. This study aims to revisit Axelrod's first tournament and incorporate noise to observe whether there is a common characteristic that consistently shows up in all successful strategies at different noise levels.

3. Materials and Methods

3.1 Tournament Framework

Axelrod's first tournament and its fifteen strategies were recreated with Python, where multiple variables could be manipulated. All strategies implemented were based on the description in the Axelrod library and "Effective Choice in Prisoner's Dilemma". The payoffs followed the convention where T = 5, R = 3, P = 1, S = 0. The original condition was repeated to verify the effectiveness of the Python model. In the original first tournament, Tit for Tat was ranked first place, but in this recreated first tournament, Nydegger won first place, and Tit for Tat placed second, as will be seen later in the paper. It is important to note this discrepancy from the original tournament, yet the general traits of dominant strategies (nice, forgiving, provokable, and clear) still held, supporting the effectiveness of the recreated model. The full code for the recreated tournament can be found at https://github.com/JustinTime0228/AxelrodNoisyIPD.

In each tournament, the fifteen strategies each played a "game" against all of the other strategies and a copy of itself. Each game consisted of 200 rounds (interactions where strategies can choose cooperation or defection), as per Axelrod's first tournament setup. The tournament was repeated ten times, and the scores were averaged to maintain validity. The tournament was then varied to increase the noise level gradually by 1%, up to 9%, and the scores were recorded. Above 10%, the tournament results became increasingly random, with strategies unable to clearly convey their meaning, so they were not included within this study. The average payoff per game and the relative rankings of each strategy were graphed to analyze the trend. (1)

3.2 Strategy Logics

The implemented strategies and general summaries of their strategic logic are listed below. The logic for the strategies was based on information available in the Axelrod Documentation Library and the original article by Professor Axelrod.

Tit for Tat: Cooperates on the first round, then copies the opponent's last move. Retaliates defection with defection, but rewards cooperation with cooperation.
Tideman and Chieruzzi: Begins with Tit for Tat, but increases the number of retaliations every time the opponent defects. The opponent is given a "fresh start" when certain conditions are met.
Nydegger: Determines next move by calculating the "point value" of the previous three moves of the opponent, giving different weight to each. When the point value equals one of the numbers in its list, it defects. Otherwise, it cooperates.
Grofman: If the opponent played a different move than Grofman, it cooperates with 2/7 probability. Otherwise, it always cooperates.
Shubik: Cooperates until the opponent defects, then defects once. For every defection by the opponent, it increases its number of retaliations by one. If the opponent defected for the second time, Shubik defects for the next two moves. It can be interpreted as similar to Tideman and Chieruzzi without the fresh start.

Stein and Rapoport: It cooperates on the first four moves and defects on the last two moves. It plays tit for tat for the rest of the game, but checks every 15 moves using the chi-squared test to see if its opponent is playing randomly. If so, it defects once.
Friedman: Starts with cooperation but defects for the remaining rounds if the opponent defects at any point.
Davis: Cooperates for the first ten rounds, then plays the same strategy as Friedman.
Graaskamp: Plays tit for tat for the first fifty rounds. It defects on the 51st round, then plays five more rounds of tit for tat. Then it checks if the opponent is playing randomly with the chi-squared test. If so, it defects for the rest of the game. It also checks if the opponent is tit for tat or a clone of itself. If so, it plays tit for tat for the remainder of the game. If not, it cooperates, but randomly defects every five to fifteen rounds.
Downing: Starts with two defections, calculates the payoff of cooperation or defection based on its history of moves. Plays the move that optimizes the payoff
Feld: Starts with tit for tat, but gradually decreases the probability of cooperation following the opponent's cooperation, starting from 1 until it is 0.5 in the 200th round.
Joss: Plays tit for tat, but it defects with 10% chance when it would usually cooperate.
Tullock: Cooperates for the first eleven moves, than randomly cooperates 10% less than the opponent on the last 10 moves
Anonymous: Originally elaborate code, yet no information about the code was released. Best interpreted as a random chance of cooperation between 30% and 70%
Random: Cooperates with 50% chance (1,5)

4. Results

4.1 Results of the Tournament

Tables 1 and 2 show the results of the tournament. The table's rows are divided by noise level, and the columns by different strategies.

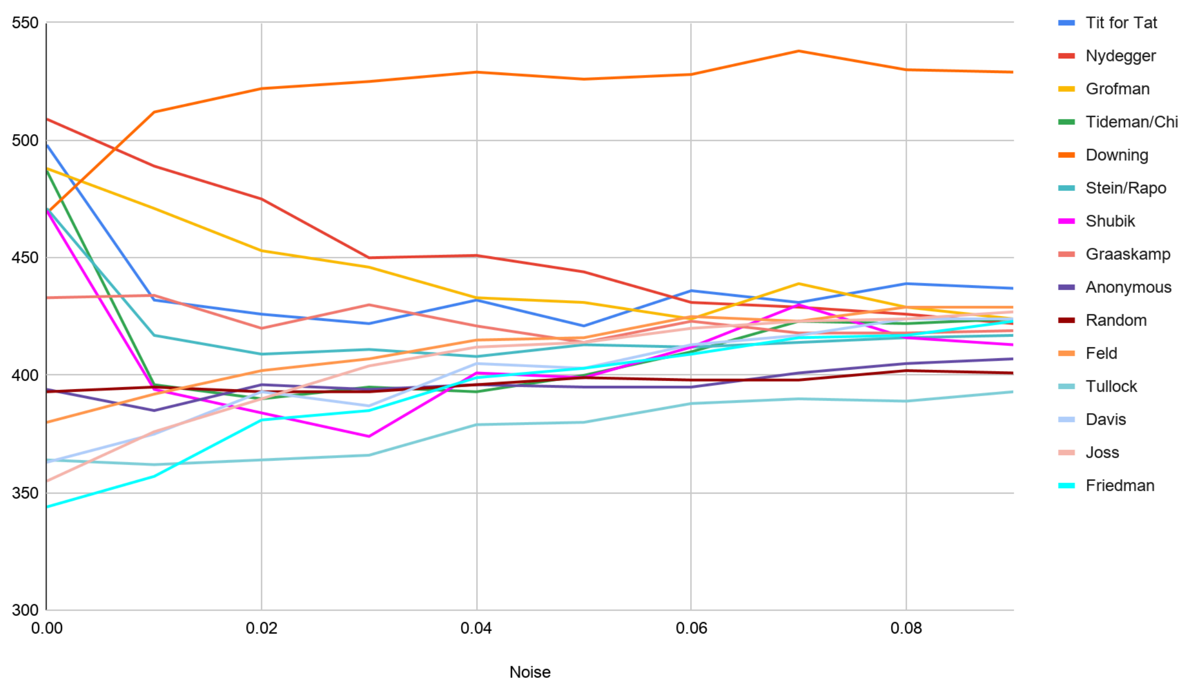
Noise	Tit for Tat	Nydegger	Grofman	Tide/Chi	Downing	Stein/Rapo	Shubik
0	498	509	488	487	469	471	470
0.01	432	489	471	396	512	417	394
0.02	426	475	453	390	522	409	384
0.03	422	450	446	395	525	411	374
0.04	432	451	433	393	529	408	401
0.05	421	444	431	400	526	413	399
0.06	436	431	424	410	528	412	412
0.07	431	429	439	423	538	414	430
0.08	439	426	429	422	530	416	416
0.09	437	422	424	424	529	417	413

Noise	Graaskamp	Anonymous	Random	Feld	Tullock	Davis	Joss	Friedman
0	433	394	393	380	364	363	355	344
0.01	434	385	395	392	362	375	376	357
0.02	420	396	393	402	364	393	390	381
0.03	430	394	393	407	366	387	404	385
0.04	421	396	396	415	379	405	412	399
0.05	414	395	399	416	380	403	414	403
0.06	423	395	398	425	388	413	420	409
0.07	418	401	398	423	390	417	423	416
0.08	418	405	402	429	389	424	424	417
0.09	419	407	401	429	393	424	427	423

4.2 Trends

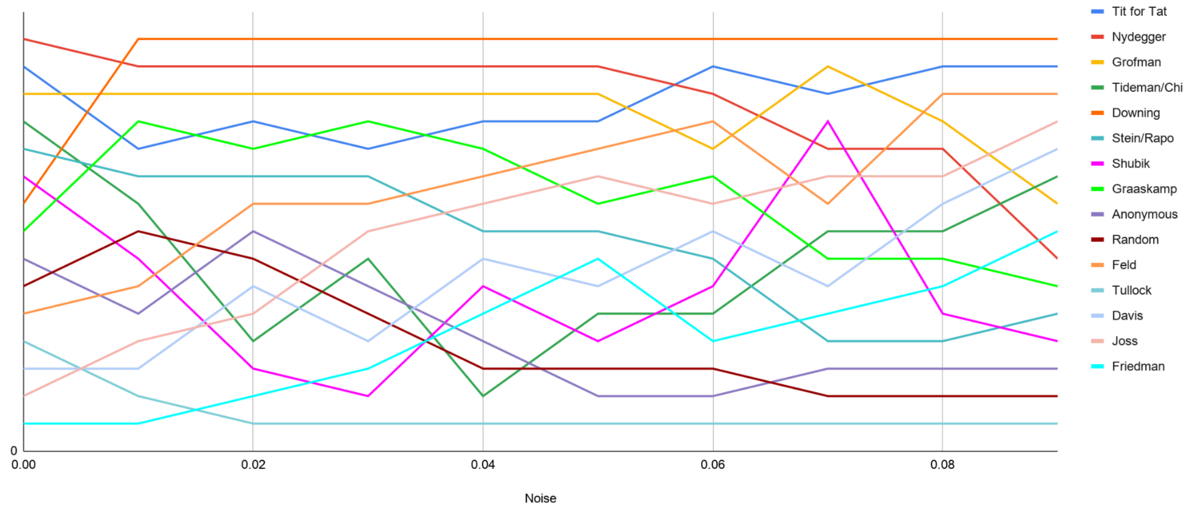
Downing was the only strategy that thrived in the increasingly noisy environment. In the original recreated tournament, with zero noise, it only ranked 7th, after six nice strategies, seen in Table 1 and Table 2.

Figure 1. Average payoffs per round of original strategies in increasingly noisy environment



However, as noise was incorporated, the other strategies' payoffs dropped significantly, converging towards the 400 - 450 point band. Downing was the only strategy that increased its own payoff in an environment with noise, and it far outscored the other strategies, as seen in Figure 1.

Figure 2. Relative rankings of the original strategies in increasingly noisy environment



The relative rankings of strategies in Figure 2 also demonstrate the unexpected results. Downing was the constant top scorer in tournaments with noise, regardless of the noise level. Tit for Tat's payoff decreased rapidly at first, but regained a firm position as the second-place strategy. The strategies that were most negatively affected by noise were Nydegger, which fell from first place to 9th place, and Stein&Rapoport, which fell from 5th place to 11th place.

5. Discussion

5.1 Flexible Strategies

It was discovered that strategies that were flexible with interpretation tended to prevail in the payoffs and rankings compared to strategies that were rigid as the noise level increased. The consistent dominance of Downing and the quick recovery of Tit for Tat suggest that flexibility is not a coincidental observation, but a defining characteristic of successful strategies.

A flexible strategy can be defined in two ways:

1. A strategy could incorporate probabilistic calculations to determine its next move, only taking into account the noisy defection as one of its many data points, thereby diminishing the impact.
2. A short-memory strategy that does not allow a defection from a round before the previous to impact its decision in the current round.

An arbitrary strategy can be identified as flexible using the following test: let the strategy play a game of Iterated Prisoner's Dilemma against any strategy, then change any cooperation to a defection. By analyzing how this change affects the behavior of the strategy, we can determine its flexibility. If it would only cause short-term changes but does not impact the long-term behavior, then it would be considered flexible.

5.2 Rigid Strategies

A rigid strategy would be the opposite of flexible, defined as follows:

A strategy that uses strict calculations that fully rely on the history of defections by the opponent to calculate its current move. Using the previously mentioned test, we can also identify rigid strategies. If the change from cooperation to defection causes long-term changes to the behavior of the strategy, then the strategy would be considered to be rigid.

5.3 Flexible and Rigid Strategies in the Tournament

Downing would fit the description of the first flexible strategy. When playing an opponent, Downing calculates the expected payoffs from each move and plays the move that could maximize its score. Therefore, even if a noisy defection was played by an opponent, it would only be included as one of Downing's data points, and the impact would be significantly diminished. Because of this, there is little probability of a chain of reciprocal defections.

Tit for Tat fits the second description of flexibility, since it has a maximally short memory. Tit for Tat only remembers what the opponent played in the last round, so even if a noisy defection was played, it would not impact the long-term play of the strategy, given that the opponent apologizes with cooperation after the noisy defection.

Nydegger is a defining example of the first category of rigid strategies. It counts the number of cooperations and defections in the previous three moves of the opponent, and calculates a score, giving different weights to each move. Then it compares the score to a list of positive integers and defects if the score matches one of these numbers. This is a very rigid strategy because it depends strictly on the opponent's history to decide on its current move instead of using a probabilistic approach. One noisy defection from the opponent can create drastically different results for Nydegger, possibly leading to more unprovoked defections. Other examples from this category are strategies that incorporate the chi-squared test, such as Graaskamp and Stein&Rapoport. The chi-squared test is used to determine if the opponent is playing randomly. If so, these strategies defect for the rest of the game. However, high noise level confounds the process, whether that be a cooperation turning into defection or vice versa. If the noise causes these strategies to falsely determine that the opponent is random, the ongoing defection would lead them to a suboptimal performance.

5.4 Success of Flexible Strategies

At lower noise levels (1-5%), this lack of flexibility does not cause rigid strategies to fail, as noise is not as prevalent as at higher noise levels, and not many noisy defections occurred. However, at higher noise levels, the noisy defections become more common, becoming a significant issue for the rigid strategies. More and more of their intentions are misinterpreted, and they are getting misinformation, which significantly affects the calculation algorithm of the rigid strategies.

5.5 Real-world Implications

This can be applied in real-world diplomatic situations with historical examples. In 1980, the US early warning system detected a missile launch from the Soviet Union, triggering a rapid defense response. However, the US suspended the alert when the alert system did not detect any further signs of attack. The alert was found to be a false alarm, caused by a faulty computer chip. (4) Furthermore, in 1983, the Soviet Union's alert system detected a US launch of an Intercontinental Ballistic Missile (ICBM). The appropriate response due to protocol was to immediately launch an ICBM to the US, as noted in the Mutually Assured Destruction doctrine. However, the officer on duty, Stanislav Petrov, questioned the validity of the alert, since it only detected one ICBM. It was expected that when the US launched ICBMs, it would launch hundreds to disarm the Soviet anti-missile defense. Petrov concluded that the alert was faulty and dismissed it until more alerts were notified. After the incident, the investigation found that sunlight's alignment with the clouds caused the alert system to mistakenly detect a missile. (9)

These two nuclear close calls are evidence of the effectiveness of flexibility for application in international relations. Both the US and the Soviet Union avoided nuclear war due to their dismissal of one-time alerts. This aligns with the logic of flexible strategies; if only one defection is noted, it is only applied as one data point among many, not as a definitive proof of aggression.

The Downing strategy is most closely associated with the current strategy used in this example. Downing's logic is to use the previous data to calculate the expected payoff to determine the best course of action. The officers on duty put the data into context of the historical data, and determined that defecting, or pressing the button for counterlaunch of nuclear weapons, would be much less likely to bring a higher payoff compared to ignoring the signal.

6. Conclusion

This study discovered the prevalence of a new type of strategy in the noisy version of the IPD. Flexibility, as a successful characteristic, can be explained in real-world terms as being less sensitive to each piece of information and instead focusing on the overall trend, seen through larger amounts of data collected over an extended period.

Yet, flexibility is predicted to have its weaknesses. If other strategies are aware of the prevalence of flexible strategies, they could potentially take advantage of the flexible strategies by probing the limit at which flexible strategies retaliate. Once this information is obtained, the opponent strategies could test the limit and only defect up to the limit, so that the flexible strategies would regard the defections as noisy and retaliate less.

Furthermore, additional research is needed to improve the model into a more realistic version. The noise of each tournament within this research is fixed, and an in-depth research into variable noise may be beneficial to create a more accurate model of international relations. It is still unknown what the noise level in real-world international relations is. It is therefore impossible, as of now, to create a model that perfectly simulates the interaction between two real-world entities. A deeper investigation into both the simulations and the real-world interactions could provide insight into the true nature of noise in international relations.

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